INCH-POUND

MS27742E
27 November 2003
SUPERSEDING
MS27742D(USAF)
31 Jan 1992

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, TYPE I, MAGNETIC LATCH, 25 AMPERES, 3 PDT, ALL WELDED, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

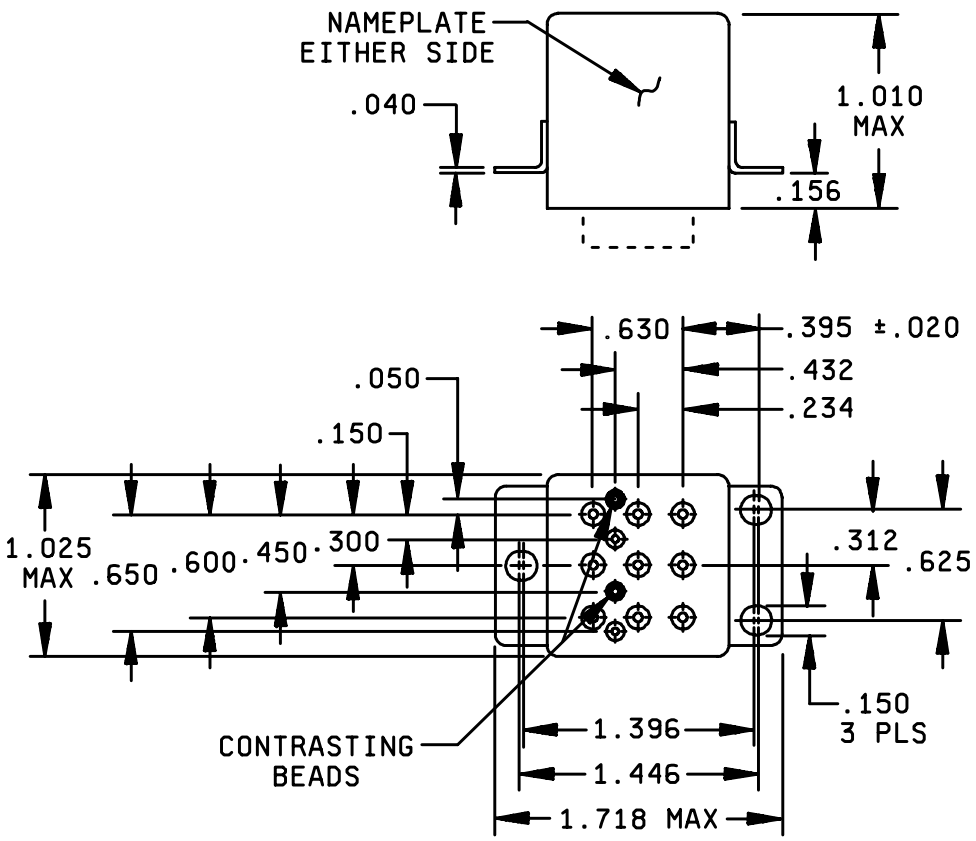
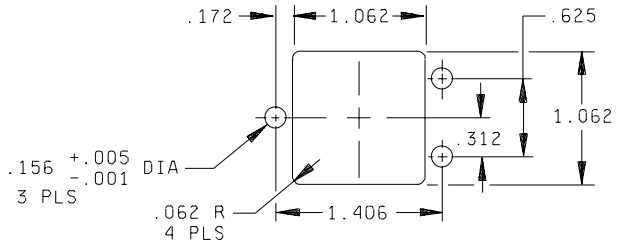


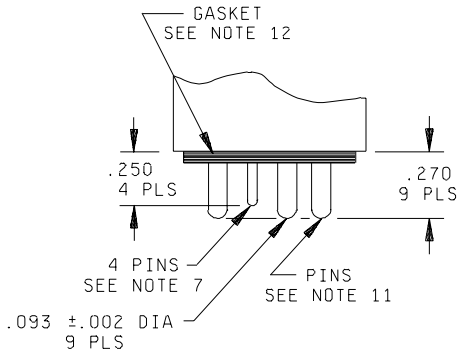

FIGURE 1. Dimensions and configuration.



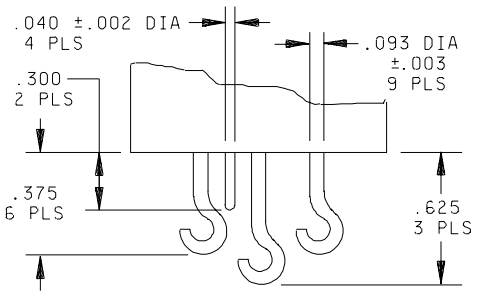
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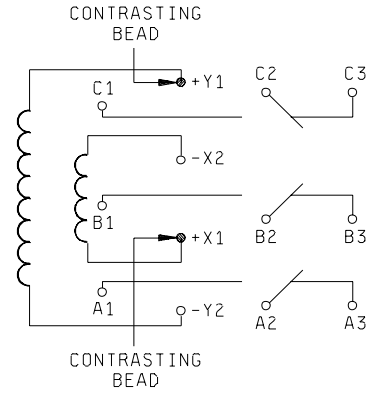

MOUNTING HOLE LAYOUT FOR -1




SOCKET PIN




SOLDER HOOK
(ALL HOOKS TIN PLATED)




TO CLOSE NO.1 CONTACTS(OPERATE)
ENERGIZE X1 AND X2
TO CLOSE NO.3 CONTACTS(RESET)
ENERGIZE Y1 AND Y2

CIRCUIT DIAGRAM
POLARIZED BISTABLE

FIGURE 1. Dimensions and configurations.- Continued



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| Inches | mm | | | Inches | mm |
|---|---|---|---|---|---|
| .001 | 0.02 | .172 | 4.36 | .600 | 15.24 |
| .002 | 0.05 | .218 | 5.53 | .625 | 15.87 |
| .003 | 0.07 | .234 | 5.94 | .630 | 16.00 |
| .005 | 0.12 | .250 | 6.35 | .650 | 16.51 |
| .020 | 0.50 | .270 | 6.85 | 1.010 | 25.65 |
| .040 | 1.01 | .300 | 7.62 | 1.025 | 26.03 |
| .050 | 1.27 | .312 | 7.92 | 1.062 | 26.97 |
| .062 | 1.57 | .375 | 9.52 | 1.396 | 35.45 |
| .093 | 2.36 | .395 | 10.03 | 1.446 | 36.72 |
| .150 | 3.81 | .432 | 10.97 | 1.718 | 43.63 |
| .156 | 3.96 | .450 | 11.43 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of specifications and standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.
7. Relay is magnetically latched in both positions.
8. Note to observe polarity must appear on relays.
9. X1 and X2 diameter .062 ±.001. Y1 and Y2 diameter .040 ±.001.
10. Coils are not to be energized simultaneously.
11. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
12. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 20 ±5, thickness .050 ±.005. Gasket material according to AMS 3332 has been considered acceptable.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Contact data:

Load ratings: See table I.

Initial contact voltage drop: 0.150 volt.

After life test: 0.175 V.

Overload current: 50 amperes dc; 80 amperes ac.

Rupture current: 60 amperes dc; 100 amperes ac.

Coil data: See table II.

Duty rating: Continuous.

Electrical data:

Insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric withstanding voltage:

Sea level:

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 | 1,000 |
| Aux contacts | N/A | N/A |
| All other points | 1,250 | 1,000 |

Altitude:

| | 80,000 ft | 300,000 ft |
|---|---|---|
| Coil to case | 350 | 500 |
| Aux contacts | N/A | |
| All other points | 350 | 500 |

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock g-level: 200 g's.

Duration: 6 ms.

Max duration contact opening: 10 µs.

Vibration - sinusoidal:

G-level 30 g's.

Frequency range 10 - 3,000 Hz.



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Vibration - random:

Applicable specification: MIL-STD-202, method 214.

Test condition: IG.

Duration: 15 minutes each plane.

Acceleration 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: 0.18 pound maximum.

Part or Identifying Number (PIN): MS27742- (dash number from table II).

Qualification by similarity: See MIL-PRF-6106.

TABLE I. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | | 400 Hz | 60 Hz | |
| Resistive 2/ | 50 | 25 | 25 | | | 25 | | | | 25 | | | | |
| Inductive | 10 | 12 | 12 | | | | | | | | | | | |
| Inductive | 20 | | | | | 15 | | | | 15 | | | | |
| Motor | 50 | 10 | 10 | | | 10 | | | | 10 | | | | |
| Lamp | 50 | 5 | 5 | | | 5 | | | | 5 | | | | |
| Transfer load | --- | | | | | --- | | | | --- | | | | 3/ |
| Mechanical life reduced current | 200 | 6 | 6 | | | 6 | | | | 6 | | | | |
| Mixed loads | Applicable per specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ For full rated load, temperature, and altitude, use number 12 wire or larger. Relays shall be mounted so that mounting bracket temperature is limited to +135°C maximum.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

TABLE II Dash number and characteristics.

| Part no. MS 27742- | Coil data | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Terminal type | Rated | | | Maximum At +25°C | | Max pick-up voltage | | | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact Bounce | | | |
| | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res | Volts | Amp | Normal 2/ | High temp test | Cont current test | | | | NO | NC | NO | NC |
| 1 | Solder hook | 28 | dc | --- | 29 | .075 | 18 | 19.8 | 22.5 | N/A | 15 | --- | 1.0 | 1.0 | --- | --- |
| 2 | Socket pin | 28 | Dc | --- | 29 | .075 | 18 | 19.8 | 22.5 | N/A | 15 | --- | 1.0 | 1.0 | --- | --- |

1/ Caution: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range..
3/ With rated coil voltage.
4/ From rated coil voltage.



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NOTES

Referenced documents. In addition to MIL-PRF-6106, this specification sheet references the following documents. (Government documents are available on line at http://assist.daps.dla.mil/quicksearch or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094). Society of Automotive Engineers documents are available from the Society of Automotive Engineers 400 Commonwealth Drive Warrendale, Pennsylvania, United States, 15096-0001. http://www.sae.org

STANDARDS

Department of Defense

MIL-STD-202 - Electronic and Electrical Component Parts

Society of Automotive Engineers (SAE)

SAE-AMS3332 - Silicone Rubber Extreme Low-Temperature Resistant, 15-30

Custodians:
Air Force - 11
DLA - CC

Review activities:
AF - 99

Preparing activity:
DLA - CC

(Project 5945-1221)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil.



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